CURVILINEAR FILTERS

A MATLAB REFERENCE MANUAL

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Summary

This manual documents the `CurvilinearFilters` package, a collection of MATLAB modules for detecting and enhancing curvilinear structures in 2-D images prior to segmentation. Five complementary filter families are covered:

Hessian Multiscale Hessian-based detectors (vesselness, ridge, blobness, plateness, neuriteness).

MFAT Multiscale Fractional Anisotropy Tensor — deterministic and probabilistic variants.

Phase Congruency Log-Gabor filter bank edge/corner detector and phase symmetry measure.

SOAGK Steerable Orientation-Aligned Gaussian Kernels, an anisotropic line detector.

Vesselness (optimised) A standalone optimised implementation of the original Frangi vessel filter, retained as a legacy reference.

All modules share a common synthetic benchmark image and a unified figure directory for reproducible documentation.

Introduction

Curvilinear structures—blood vessels, neurites, fibres, and reticular networks—are ubiquitous in biological and industrial imaging. Their reliable detection is a prerequisite for segmentation, tracing, and quantification pipelines. No single filter is universally optimal: tubular structures, ridge lines, and symmetric features each call for different mathematical characterisations of local image geometry.

This package provides a curated set of filters that together span the major design strategies: second-order differential analysis (Hessian family), tensor-based fractional anisotropy (MFAT), Fourier-domain phase measurement (Phase Congruency/Symmetry), and steerable filter banks (SOAGK). Each module is implemented in clean, tested MATLAB with no toolbox dependencies beyond the Image Processing Toolbox.

Mathematical Background

The Hessian Matrix and Scale Space

Given a greyscale image $I(x, y)$, the Hessian matrix at scale σ is

$$\mathbf{H}_\sigma = \sigma^2 \begin{pmatrix} \partial_{xx} I_\sigma & \partial_{xy} I_\sigma \\ \partial_{xy} I_\sigma & \partial_{yy} I_\sigma \end{pmatrix}, \tag{1}$$

where $I_\sigma = I * G_\sigma$ denotes Gaussian smoothing and the σ^2 factor provides Lindeberg's scale normalisation, enabling meaningful cross-scale comparison of responses.

Let λ_1, λ_2 be the eigenvalues ordered by $|\lambda_1| \leq |\lambda_2|$. Their signs and ratio characterise local structure:

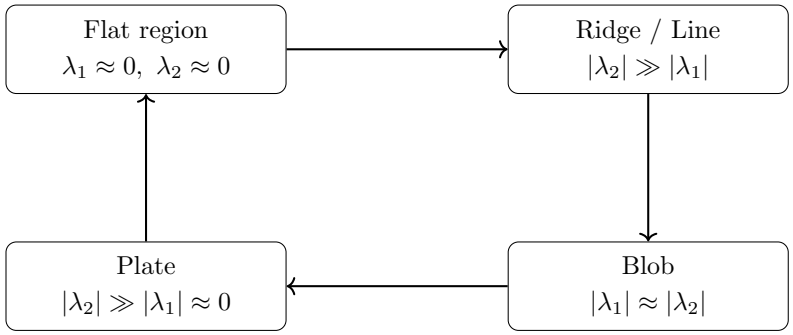

Figure 1: Interpretation of Hessian eigenvalues in 2D: flat regions ($|\lambda_1| \approx |\lambda_2| \approx 0$), ridges ($|\lambda_2| \gg |\lambda_1|$, $\lambda_2 < 0$), and blobs ($|\lambda_1| \approx |\lambda_2|$, both negative).

The eigenvector corresponding to λ_2 points across a ridge; the tangent direction is obtained by a 90° rotation.

Log-Gabor Filters and Phase Congruency

Log-Gabor filters are band-pass filters constructed in the frequency domain:

$$G(f) = \exp\left(\frac{-(\ln(f/f_0))^2}{2(\ln(\sigma_f))^2}\right), \tag{2}$$

where f_0 is the centre frequency and σ_f controls bandwidth. A bank of log-Gabor filters at multiple scales and orientations decomposes the image into oriented frequency sub-bands. Phase congruency measures the degree to which these components are in phase at each location, providing an illumination-invariant feature measure.

Fractional Anisotropy

In diffusion tensor imaging, fractional anisotropy (FA) quantifies the degree to which diffusion is directionally asymmetric. MFAT adapts this concept to image Hessian eigenvalues: a high FA-like score indicates a strongly curvilinear (anisotropic) local geometry, independent of contrast magnitude.

Hessian-Based Filters

Overview

The Hessian module provides five filter types sharing a common low-level numerical core (`applyHessian2D`, `eig2image`) and accessed through a single entry-point function `hessian2DFilters`.

API

```
[R, scale, direction] = hessian2DFilters(I, ...)
% 'FilterType' 'vesselness' | 'ridge' | 'blob' | 'plate'
```

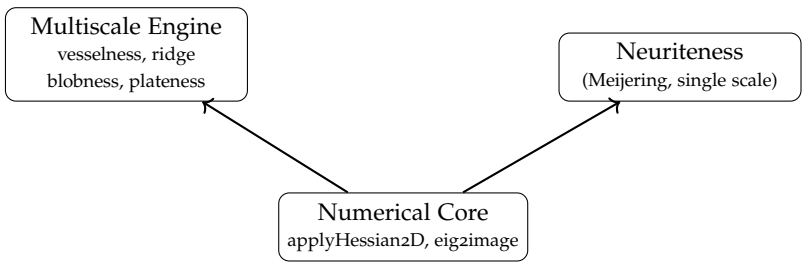

Figure 2: Software architecture of the Hessian module. The numerical core is shared between the multiscale engine and the single-scale neuriteness filter.

```
% 'Sigmas' vector of scales (default: auto-computed)
% 'WhiteOnDark' true | false (default: true)
% 'Precision' 'single' | 'double' (default: 'single')
% 'Parameters' struct with filter-specific overrides

[N, direction] = neuriteness2D(I, sigma)
```

Vesselness (Frangi, 1998)

The vesselness response at scale σ combines Hessian eigenvalues into a score that rewards elongated, tubular geometry:

$$\mathcal{R}_B = \frac{\lambda_1}{\lambda_2}, \tag{3}$$

$$S^2 = \lambda_1^2 + \lambda_2^2, \tag{4}$$

$$V_\sigma = \exp\left(\frac{-\mathcal{R}_B^2}{2\beta^2}\right)\left(1 - \exp\left(\frac{-S^2}{2c^2}\right)\right), \tag{5}$$

with defaults $\beta = 0.5$, $c = 15$. The final response is $V(x) = \max_\sigma V_\sigma(x)$.

Ridge Filter (Sato, 1998)

The ridge response rewards line-like structures by penalising blob-like responses via the ratio λ_1/λ_2.

Blobness (Lindeberg, 1998)

Blobness uses the normalised determinant of the Hessian: $B = \lambda_1\lambda_2$.

Plateness

The 2-D plateness measure detects thick elongated regions with low curvature anisotropy, analogous to the plate measure used in 3-D analysis.

Neuriteness (Meijering et al., 2004)

Modified eigenvalues are formed as $\tilde{\lambda}_1 = \lambda_1 + \alpha\lambda_2$, $\tilde{\lambda}_2 = \lambda_2 + \alpha\lambda_1$, with $\alpha = -1/3$, then reordered by magnitude. The response

$$N(x) = \begin{cases} \tilde{\lambda}_1(x)/\min_x(\tilde{\lambda}_1), & \tilde{\lambda}_1(x) < 0, \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

yields a normalised, contrast-invariant, single-scale measure in $[0, 1]$.

Results

MFAT — Multiscale Fractional Anisotropy Tensor

Overview

MFAT adapts the fractional anisotropy (FA) concept from diffusion MRI to 2-D image analysis [1]. Two complementary outputs are provided:

[1]

MFAT-λ Deterministic backbone. Produces a near-binary evidence map via strict post-processing rules.

MFAT-Prob Probabilistic extension. Accumulates Beta-distributed likelihoods across scales and returns a posterior probability in $[0, 1]$.

Optional modifiers (`mfatEntropyWeight`, `mfatFractional`) adjust the response for uncertain or weak-signal regions.

API

```
out = mfatLambda(I, sigmas, ...)
% 'tau' lower eigenvalue clipping factor (default: 0.03)
% 'tau2' upper eigenvalue clipping factor (default: 0.30)
% 'D' aggregation strength (default: 0.27)
% 'whiteOnDark' true | false (default: true)
% 'precision' 'single' | 'double' (default: 'single')

out = mfatProb(I, sigmas, ...) % same parameters
```

Algorithm: MFAT-λ

For each scale σ:

1. Compute scale-normalised Hessian eigenvalues λ_1, λ_2.
2. Apply three-factor regularisation (clipping at τ and τ_2).
3. Compute the FA-like tensor measure:

$$\mathrm{FA} = \frac{|\lambda_2 - \lambda_1|}{\sqrt{\lambda_1^2 + \lambda_2^2 + \varepsilon}}. \tag{7}$$

4. Apply strict post-processing: response $\leftarrow 0$ if either eigenvalue ≥ 0; aggregate via $D \tanh(\cdot - D) + \max$ fusion.

Algorithm: MFAT-Prob

For each scale σ, model the MFAT-λ response via Beta likelihoods:

$$p_{\mathrm{vessel}}(r) = \mathrm{Beta}(r; 2, 1), \tag{8}$$

$$p_{\mathrm{bg}}(r) = \mathrm{Beta}(r; 1, 2). \tag{9}$$

Accumulate log-likelihood ratios across scales, then convert via a logistic transform: $P(x) = (1 + \exp(-\mathrm{LLR}(x)))^{-1}$.

Results

Phase Congruency and Phase Symmetry

Overview

Phase congruency [2] measures feature strength as the degree to which log-Gabor sub-bands are in phase, making it invariant to image contrast. Phase symmetry extends this to detect symmetric (line-like and blob-like) patterns. Both are implemented as optimised MATLAB functions with persistent filter caches for efficiency.

[2]

API

```
[M, m, or, featType, PC, EO, T, pcSum, noiseMask] = ...
   phaseCong3Optimised(I, ...)
% 'nscale' number of scales (default: 4)
% 'norient' number of orientations (default: 6)
% 'minWaveLength' smallest scale wavelength (default: 5)
% 'mult' scaling between successive filters (default: 2.1)
% 'sigmaOnf' s.d.-to-centre-frequency ratio (default: 0.55)
% 'k' noise threshold parameter (default: 2.0)
% 'precision' 'single' | 'double' | 'like' (default: 'double')

[phaseSym, orientation, totalEnergy, T] = phaseSymOptimised(I,
     ...)
% 'polarity' 1 (bright) | -1 (dark) | 0 (both, default)
```

Phase Congruency

At each point x, PC is the weighted ratio of aligned energy to total filter response energy across all scales and orientations. Key outputs:

M Maximum moment of the PC covariance matrix (edge strength).

m Minimum moment (corner/junction strength).

or Local orientation, in degrees $[0, 180)$.

Phase Symmetry

Phase symmetry sums the weighted phase alignment across orientations for each scale, with polarity control allowing detection of bright, dark, or both types of symmetric features.

Results

SOAGK — Steerable Orientation-Aligned Gaussian Kernels

Overview

SOAGK implements a steerable anisotropic Gaussian line detector. A basis of second-order anisotropic Gaussian derivative kernels is computed once per scale/anisotropy pair and then steered analytically across orientations, avoiding redundant convolutions.

API

```
[lineMap, dirMap] = AGlineDetectorSteerableConv2( ...
   im, sigmas, thetas, rhos, precision)
% im grayscale image in [0, 1]
% sigmas vector of scales
% thetas orientations in [0, pi)
% rhos anisotropy factors >= 1
% precision 'single' | 'double' (default: 'single')
%
% lineMap line strength in [0, 1]
% dirMap dominant orientation (radians)
```

An `imfilter`-based variant (`AGlineDetectorSteerable`) is also provided and produces numerically equivalent results.

Algorithm

1. For each (σ, ρ) pair, construct the steerable basis $\{G_{xx}, G_{yy}, G_{xy}\}$ of anisotropic second-order Gaussian derivatives in the stretched coordinate frame $(x' = \rho x,\ y' = y/\rho)$.
2. Convolve the image with each basis kernel once.
3. For each orientation θ, steer the response:

$$R_\theta = \cos^2\theta\ (G_{xx} * I) + \sin^2\theta\ (G_{yy} * I) + 2\cos\theta\sin\theta\ (G_{xy} * I). \quad (10)$$

4. Retain $\max_{\sigma,\rho,\theta} R_\theta$ at each pixel and normalise to $[0, 1]$.

Results

Vesselness (Optimised Standalone)

`Vesselness/src/vesselnessFilter2DOptimised.m` is an optimised standalone implementation of the original Frangi vessel filter, retained as a legacy reference. It implements the same Frangi (1998) algorithm as the `'vesselness'` mode of `hessian2DFilters` but with a streaming-max architecture (no 3-D scale stacks) and explicit precision control.

```
[outIm, whatScale, Direction] = ...
   vesselnessFilter2DOptimised(I, options)
% options.FrangiScaleRange [min max] (default: [1 10])
% options.FrangiScaleRatio (default: 2)
% options.FrangiBetaOne (default: 0.5)
% options.FrangiBetaTwo (default: 15)
% options.WhiteOnDark (default: true)
% options.Precision (default: 'single')
```

Note: For new work, `hessian2DFilters` with `'FilterType','vesselness'` is the recommended entry point, as it provides a consistent API across all filter types and integrates auto-scale selection via `autoLearnHessianScales`.

Synthetic Benchmark Phantom

All demos share a common synthetic image generated by `makeReticulateTestImage(256, 256, 42)` in `Helper functions/`. The function produces:

- a smooth background from a seeded random field,
- 40 random nodes connected to their 3 nearest neighbours, forming a reticulate (network) structure,
- line widths of 1–3 pixels, box-filter softened,
- mild Gaussian noise ($\sigma = 0.03$).

The deterministic seed (42) ensures bit-identical images across runs and platforms. Using a shared image allows direct qualitative comparison of all filter responses.

Comparative Summary

Module	Primary target	Response	Scale	Key parameters
Hessian/vesselness	Tubular structures	Soft [0,1]	Multi	β, c
Hessian/ridge	Ridge lines	Soft [0,1]	Multi	α
Hessian/blobness	Isotropic blobs	Soft [0,1]	Multi	—
Hessian/neuriteness	Curvilinear shape	Normalised	Single	—
MFAT-λ	Curvilinear (strict)	Near-binary	Multi	τ, D
MFAT-Prob	Curvilinear (soft)	Probability	Multi	Beta prior
Phase Congruency	Edges & corners	Invariant	Multi	k, g
Phase Symmetry	Symmetric features	Invariant	Multi	polarity
SOAGK	Line segments	Normalised	Multi	ρ
Vesselness (opt.)	Tubular (legacy)	Soft [0,1]	Multi	β, c

Table 1: Summary of filter families, their primary targets, response type, scale handling, and key tuning parameters.

Test Suite

Each module ships with a unit test suite exercising documented mathematical guarantees. Tests are run with `runtests('test_<module>')` in MATLAB.

Module	Test file	Key invariants
Hessian	`test_vesselness_basic.m`	Line detection, polarity
	`test_eig2image.m`	Eigenvalue ordering
	`test_hessianEigen2D_scale.m`	Scale normalisation
	`test_hessian_direction.m`	Orientation accuracy
	`test_neuriteness_*.m` (3 files)	Blob suppression, continuity
MFAT	`TestMfat.m`	Lambda/Prob output ranges
PhaseCongruency	`test_phasecong3Optimised.m`	Output ranges, determinism, precision, cache
SOAGK	`test_AGlineDetector.m`	Size, [0,1] range, line response, orientation, conv2/imfilter equivalence
Vesselness	`test_vesselnessFilter2DOptimised.m`	Numerical agreement with reference

Table 2: Unit test coverage per module. No test asserts behaviour not guaranteed by the underlying theory.

Repository Layout and Build Instructions

Folder Structure

```
CurvilinearFilters_sandbox/
|-- Hessian/
|   |-- src/  engine/  neuriteness/  utils/
|   |-- demos/
|   `-- tests/  utilities/
|-- MFAT/
|   |-- src/  core/  drivers/  responses/  modifiers/  utils/
|   |-- config/
|   |-- demos/
|   `-- tests/
|-- PhaseCongruency/
|   |-- src/
|   |-- utils/
|   |-- demos/
|   `-- tests/
|-- SOAGK/
|   |-- src/
|   |-- utils/
|   |-- demos/
|   `-- tests/
|-- Vesselness/
|   |-- src/
|   |-- +vesselness/
|   |-- demos/
|   `-- tests/
|-- Helper functions/
|   `-- makeReticulateTestImage.m
`-- manual/
    |-- CurvilinearFilters.tex
    |-- figures/
    `-- demos/
        `-- demo_generateManualFigures.m
```

Generating Figures

Before compiling the manual, run the master figure-generation script from MATLAB:

```
cd manual/demos
run demo_generateManualFigures
```

This script adds all module paths, runs each filter on the shared test image, and saves PDF figures to `manual/figures/`.

Compiling the Manual

```
cd manual
```

```
pdflatex CurvilinearFilters.tex
pdflatex CurvilinearFilters.tex   % second pass for TOC
```

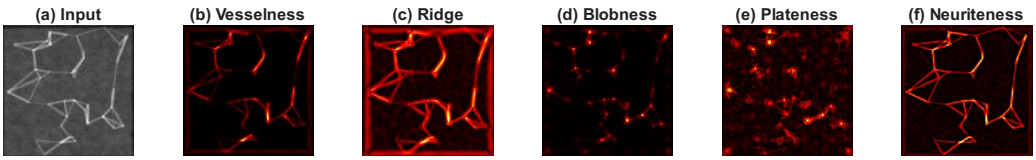

Figure 3: Hessian module responses on the reticulate benchmark image. (a) Input. (b) Vesselness. (c) Ridge. (d) Blobness. (e) Plateness. (f) Neuriteness. Observe that vesselness and ridge emphasise fine tubular segments, while neuriteness suppresses blob-like junction pixels.

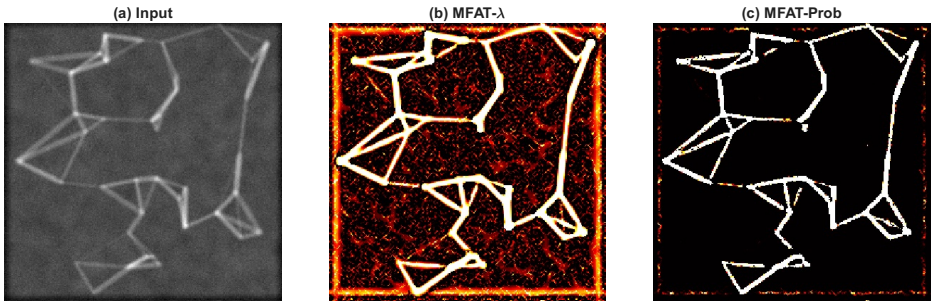

Figure 4: MFAT responses on the reticulate benchmark image. (a) Input. (b) MFAT-λ (near-binary deterministic map). (c) MFAT-Prob (posterior probability).

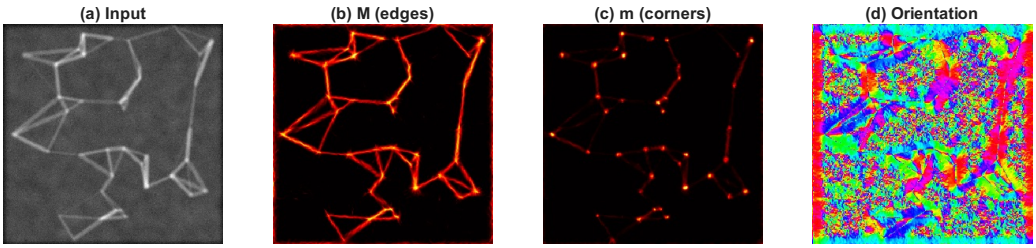

Figure 5: Phase congruency on the reticulate benchmark image. (a) Input. (b) M (edge map). (c) m (corner/junction map). (d) Orientation (HSV colourmap).

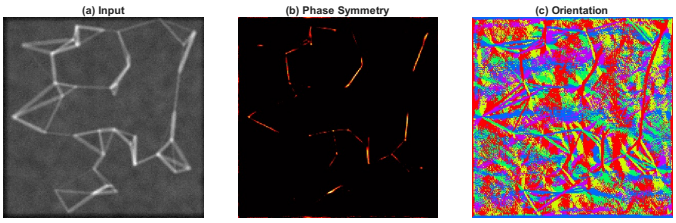

Figure 6: Phase symmetry on the reticulate benchmark image. (a) Input. (b) Symmetry response. (c) Orientation.

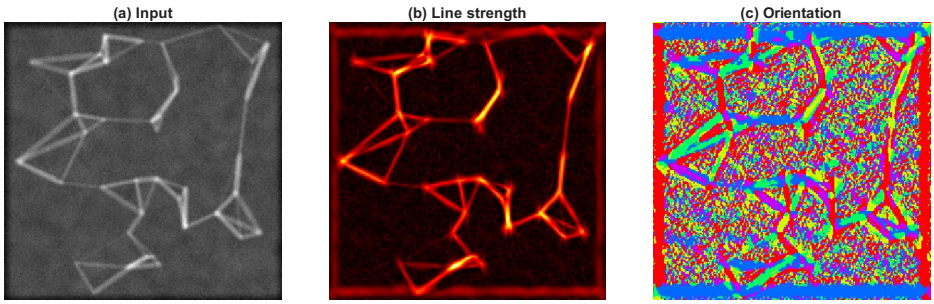

Figure 7: SOAGK line detector on the reticulate benchmark image. (a) Input. (b) Line strength map. (c) Dominant orientation (HSV colourmap).

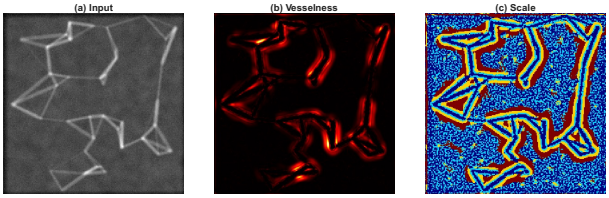

Figure 8: Optimised standalone vesselness on the reticulate benchmark image. (a) Input. (b) Vesselness response. (c) Scale index at maximum response.

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